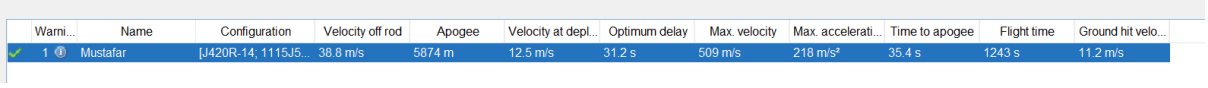
# Mission Summary

| Requirement | Mission requirement | Simulated result | Met |
|---|---|---|---|
| Apogee | 4000 ± 100 m (3900–4100 m) | 3,996.4 m at t = 32.46 s | Yes |
| Payload mass | 1.00 ± 0.05 kg | 1.00 kg mass component | Yes |
| Rocket length | ≤ 2200 mm | 2,050 mm | Yes |
| Motor total impulse | ≤ 2800 N·s combined | about 1,065 N·s combined (J420R + I285R) | Yes |
| Drogue deployment | At apogee | 32.508 s (apogee at 32.507 s) | Yes |
| Payload deployment | At apogee (4000 m) | Payload parachute opens at 32.508 s, 3,996 m | Yes |
| Main chute deployment | At apogee (4000 m) | 32.508 s, 3,996 m | Yes |
| Recovery | Parachute recovery, mandatory | Sustainer and payload under three parachutes, booster under its own | Yes |
| Descent velocity | 4–8 m/s | Sustainer and payload 6.25–6.42 m/s, booster 5.7 m/s | Yes |
| Total mass (cardboard) | ≤ 10 kg including motors | 3.78 kg at liftoff | Yes |
| Mass overrides | Not allowed | None used | Yes |

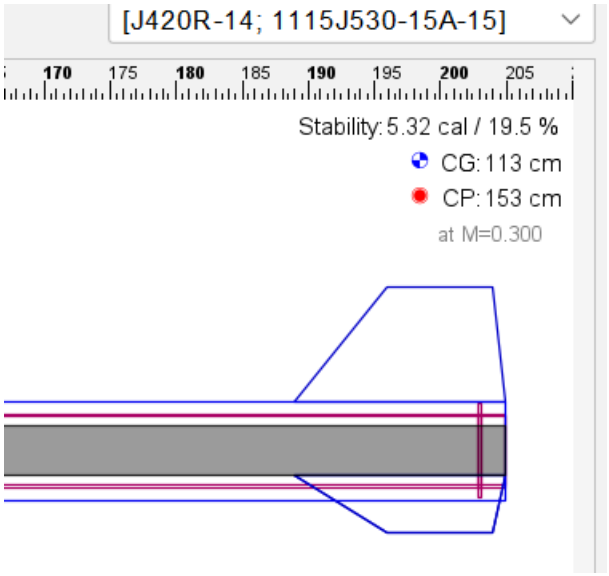
Stability stats are explained in later section

# Thought process

We aimed to pass the altitude check first and then check all the other requirements but then at first the rocket was too stable(over 5 calibers) and was reaching an apogee too high (around 5500m)



| Warni... | Name | Configuration | Velocity off rod | Apogee | Velocity at depl... | Optimum delay | Max. velocity | Max. accelerati... | Time to apogee | Flight time | Ground hit velo... |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Mustafar | [J420R-14; 1115J5... | 38.8 m/s | 5874 m | 12.5 m/s | 31.2 s | 509 m/s | 218 m/s² | 35.4 s | 1243 s | 11.2 m/s |




so to tackle this, we switched to less powerful motors and lower fin spans

# Rocket Description

| Stages | Component | Key dimensions |
|---|---|---|
| Sustainer (1,350 mm) | Nose cone | Ogive, 300 mm long 2 mm wall |
| | Payload bay | Body tube 250 mm, 1.5 mm wall; 1.00 kg payload mass component; bulkhead; payload parachute |
| | Recovery bay | Body tube 350 mm; drogue, main parachute and 1.5 m shock cord |
| | Motor bay | Body tube 450 mm; 38 mm motor mount and two centering rings |
| | Fins | 3 trapezoidal; root 140 mm, tip 62 mm, span 55 mm, sweep 50 mm, 3 mm thick |
| Booster (700 mm) | Booster bay | Body tube 700 mm; bulkhead, booster parachute, 1.0 m shock cord |
| | Motor mount | 38 mm motor in a 56 mm motor mount and two centering rings holding it |
| | Fins | 3 trapezoidal; root 160 mm, tip 80 mm, span 60 mm, sweep 70 mm, 3 mm thick |

| Stage | Motor | Simulated total impulse | Ignition | Ejection delay |
|---|---|---|---|---|
| Booster | AeroTech J420R | about 651 N·s | At launch | 14 s |
| Sustainer | AeroTech I285R | about 414 N·s | 1.3 s after booster burnout | 28 s |
| Combined | | about 1,065 N·s (limit 2,800 N·s) | | |

## Staging sequence and PLOTS

0 s: launch, booster ignites.

0.29 s: The rocket leaves the 3.5 m launch rod at 31 m/s.

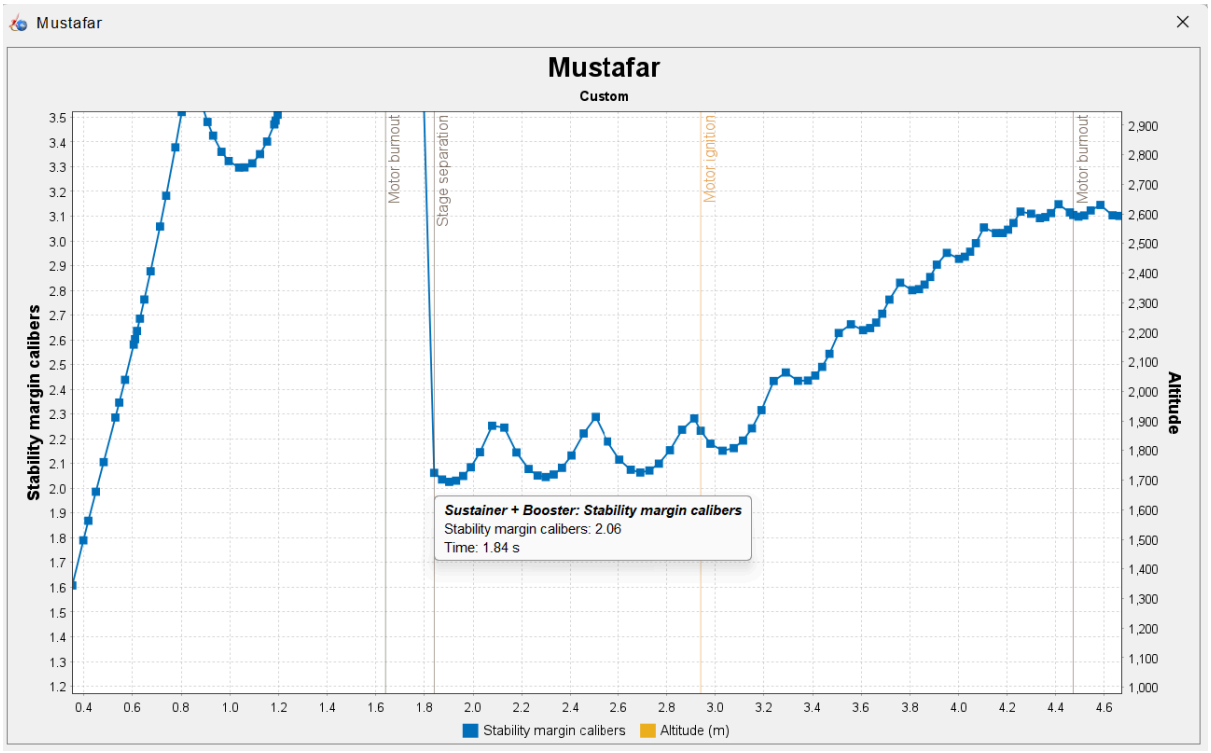
1.64 s: Booster burnout.

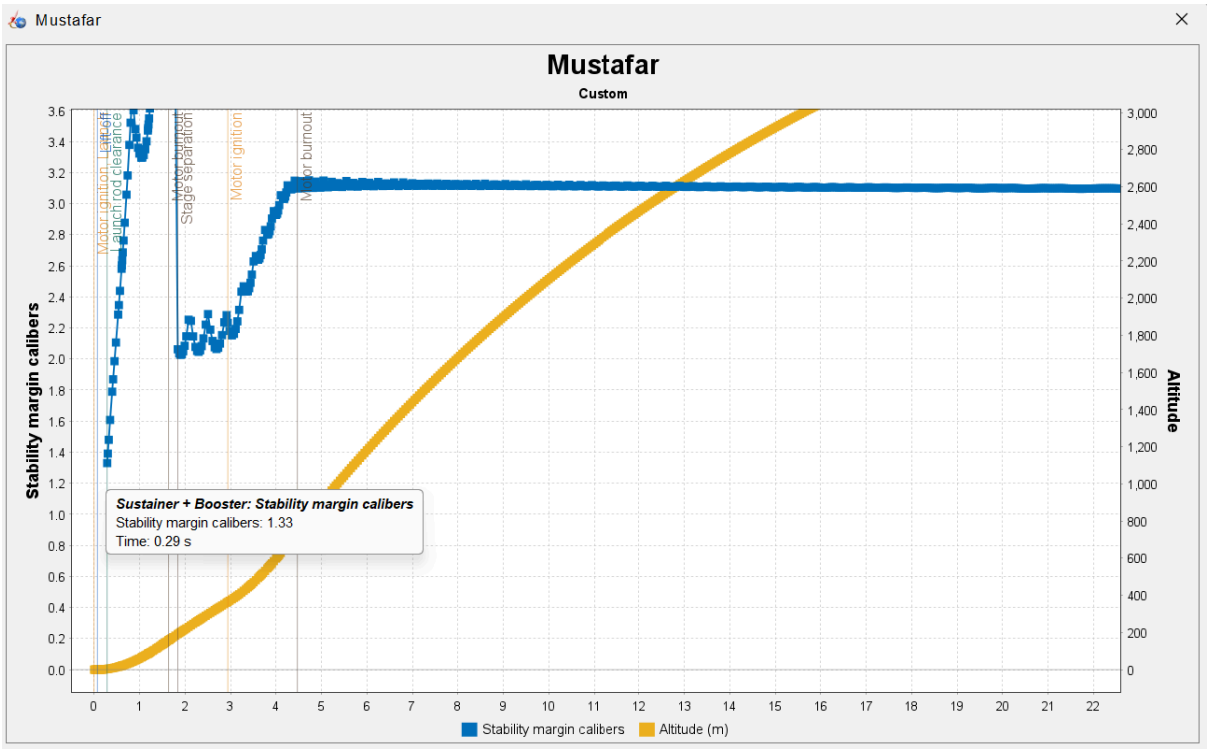
1.84 s: booster separates (0.2 s after burnout) and parachute opens

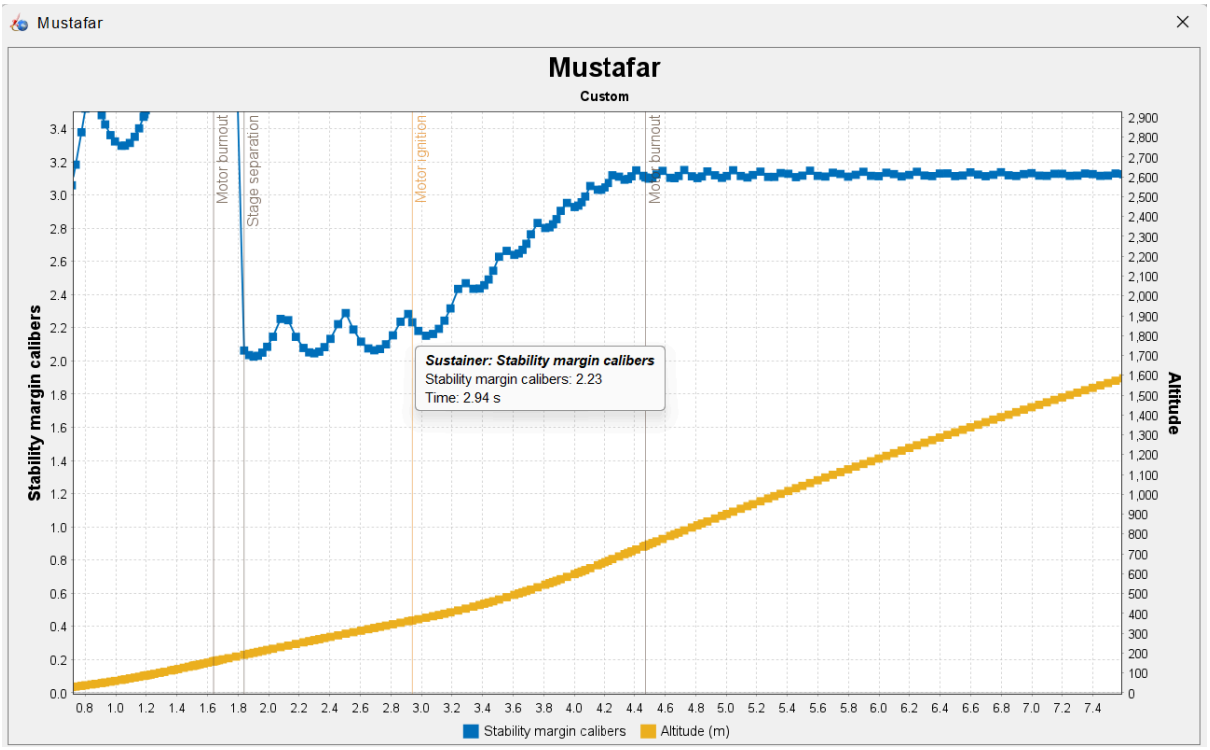
2.94 s: Sustainer ignites; burnout at 4.47 s.

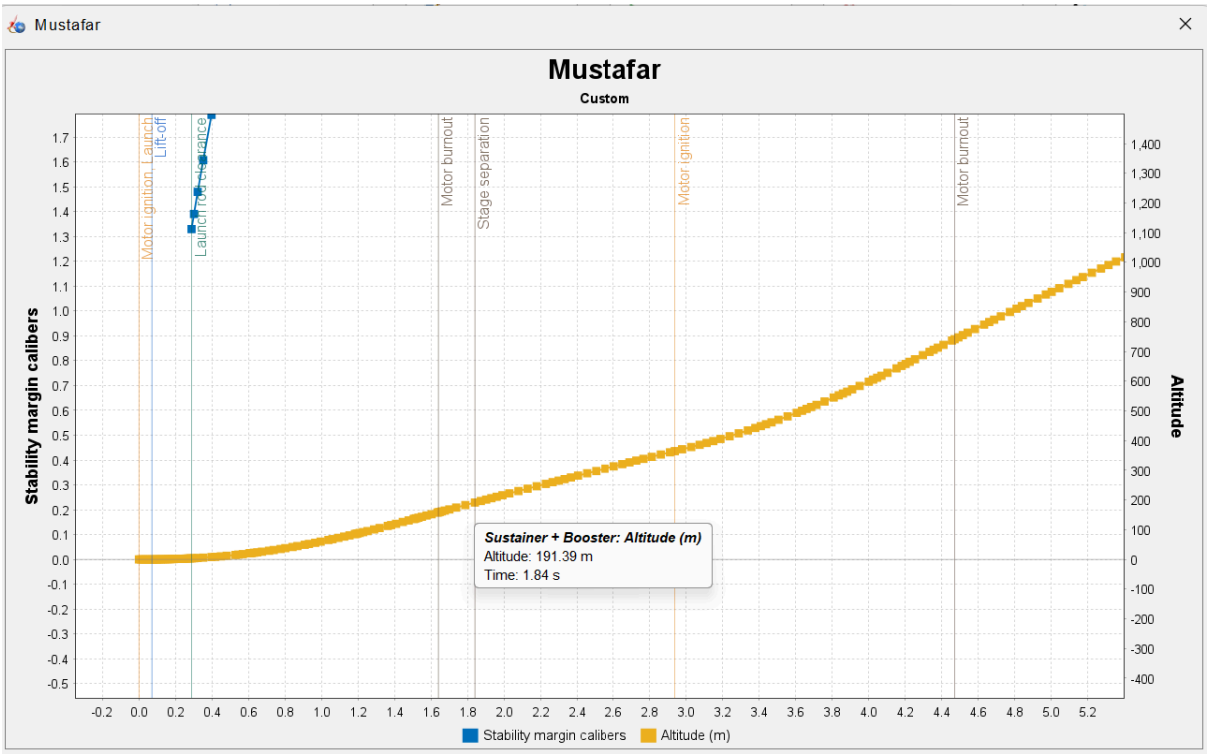
32.51 s: apogee is reached, drogue, payload and main parachutes deploys

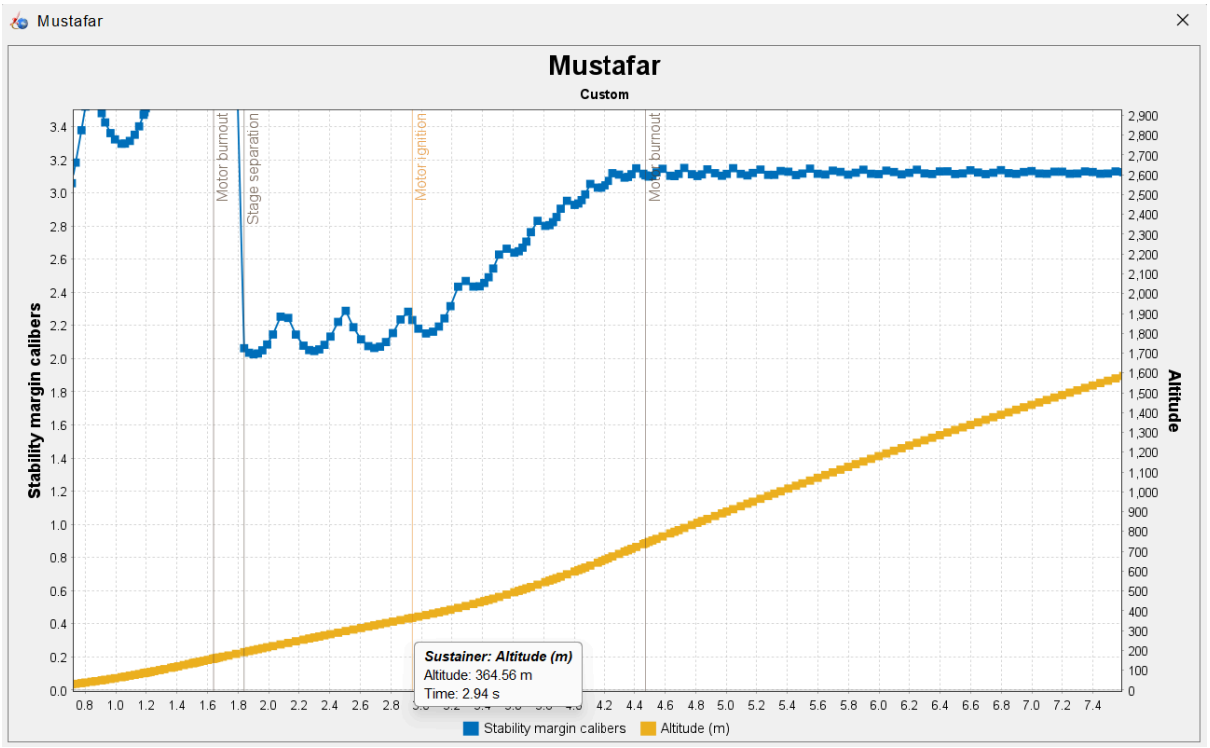
# PLOTS

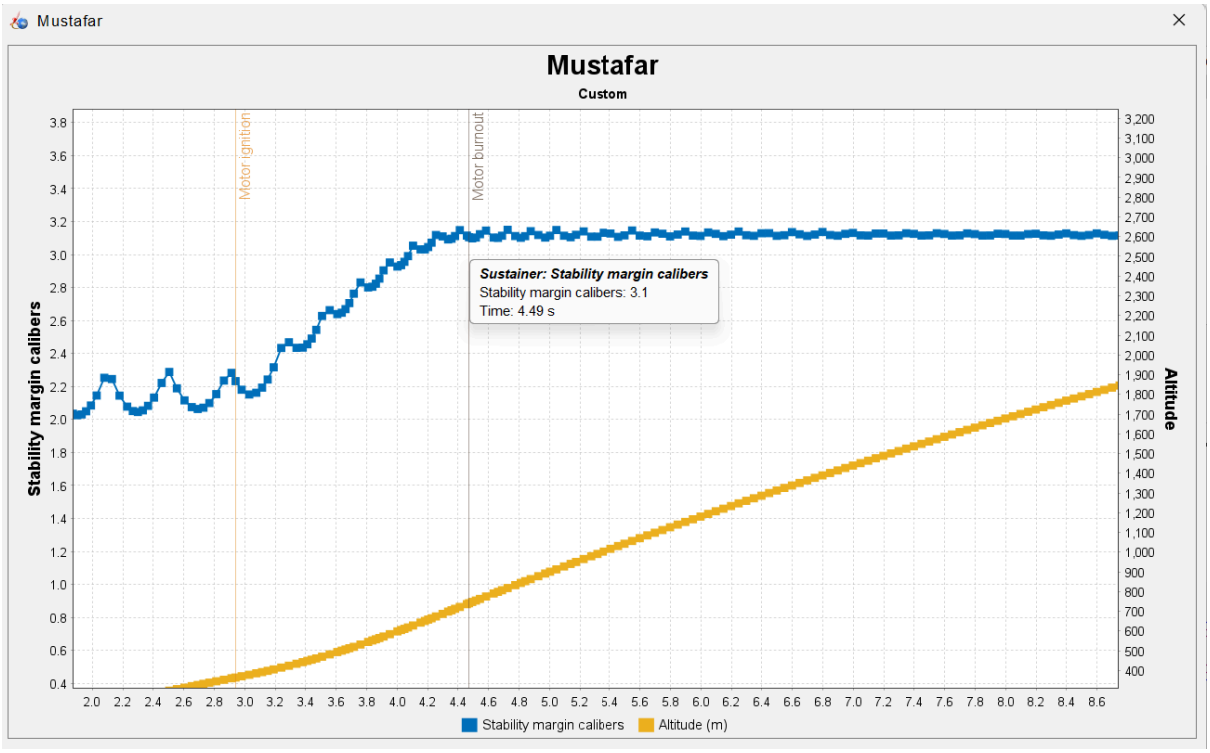

Mustafar
Mustafar
Custom
Stability margin calibers
Altitude
Motor burnout
Stage separation
Motor ignition
Motor burnout
Sustainer + Booster: Stability margin calibers
Stability margin calibers: 2.06
Time: 1.84 s
Stability margin calibers
Altitude (m)

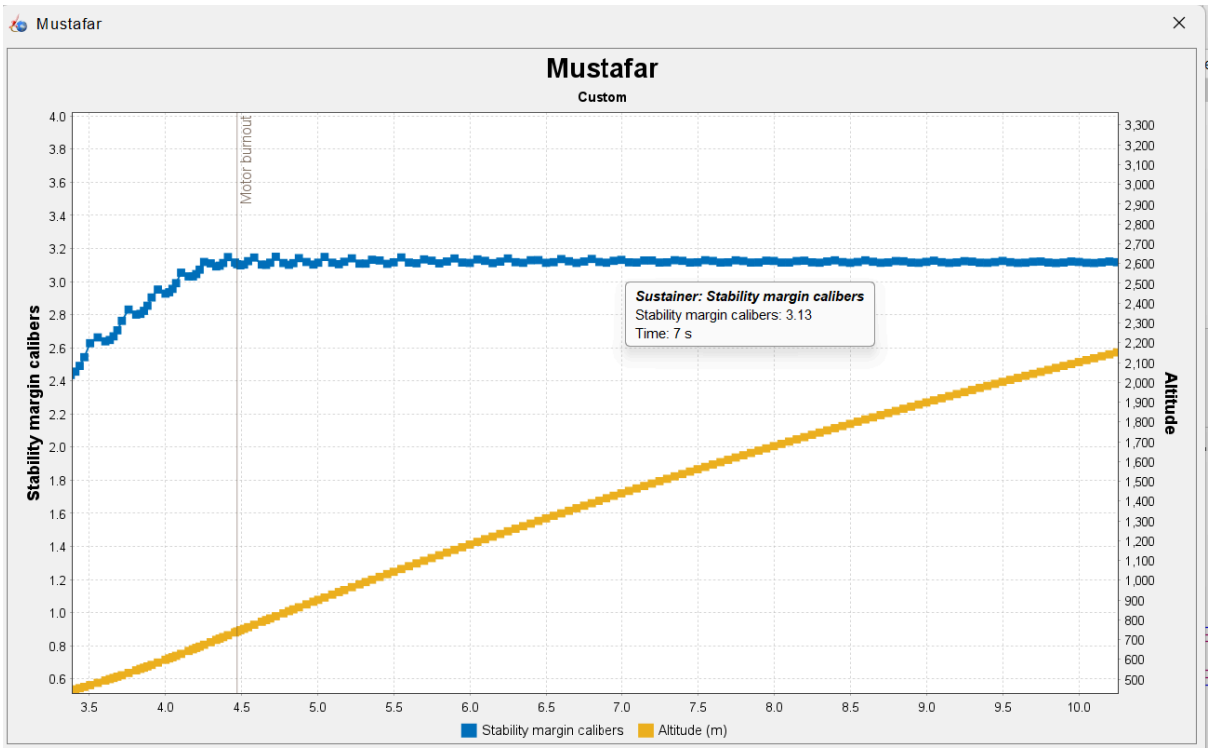

Mustafar
Mustafar
Custom
Stability margin calibers
Altitude
Motor ignition
Launch rod clearance
Motor burnout
Stage separation
Motor ignition
Motor burnout
Sustainer + Booster: Stability margin calibers
Stability margin calibers: 1.33
Time: 0.29 s
Stability margin calibers
Altitude (m)

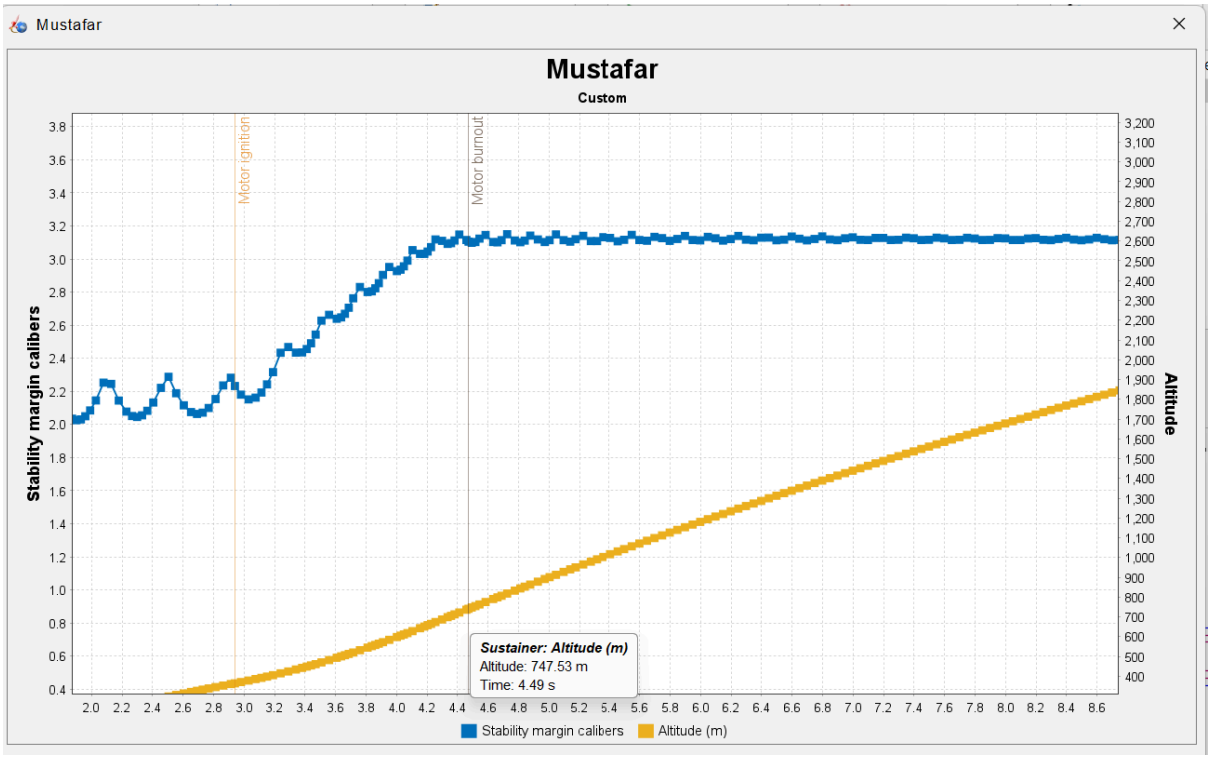

Mustafar
Mustafar
Custom
Motor burnout
Stage separation
Motor ignition
Motor burnout
Stability margin calibers
Altitude
Sustainer: Stability margin calibers
Stability margin calibers: 2.23
Time: 2.94 s
Stability margin calibers
Altitude (m)

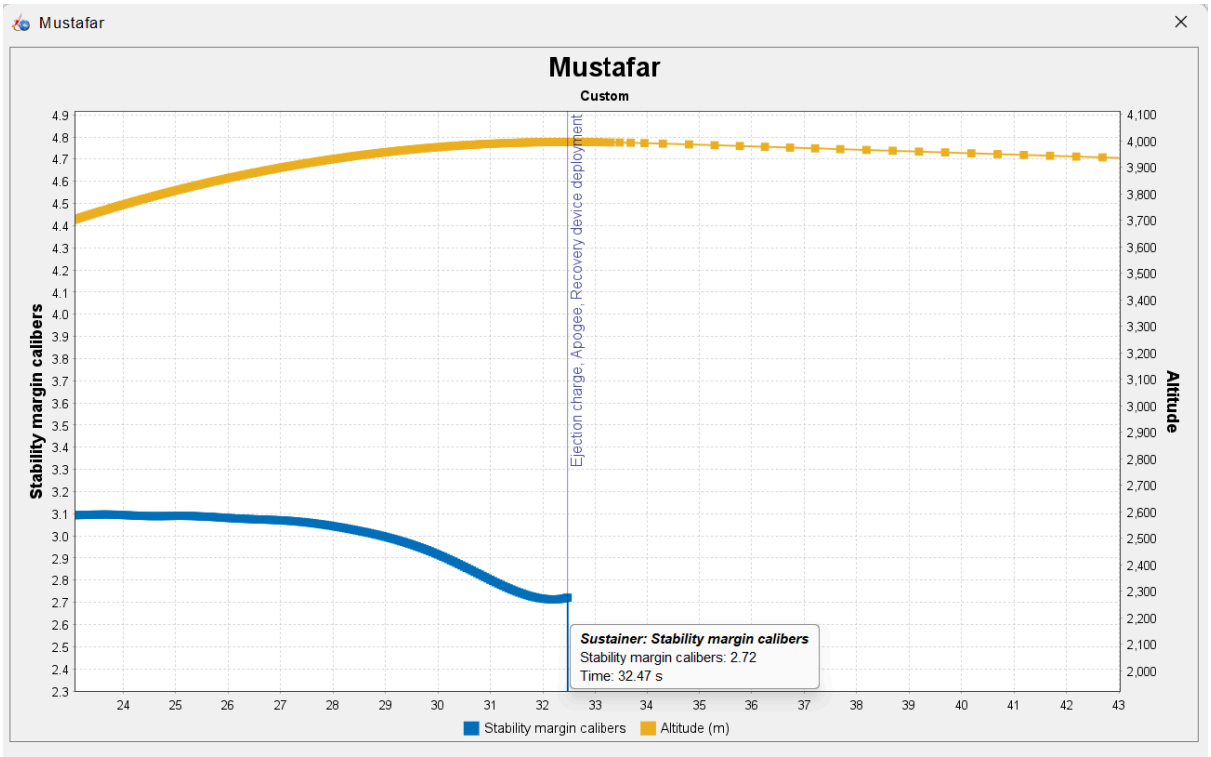

Mustafar
Mustafar
Custom
Motor ignition
Launch
Lift-off
Launch clearance
Motor burnout
Stage separation
Motor ignition
Motor burnout
Stability margin calibers
Altitude
Sustainer + Booster: Altitude (m)
Altitude: 191.39 m
Time: 1.84 s
Stability margin calibers
Altitude (m)

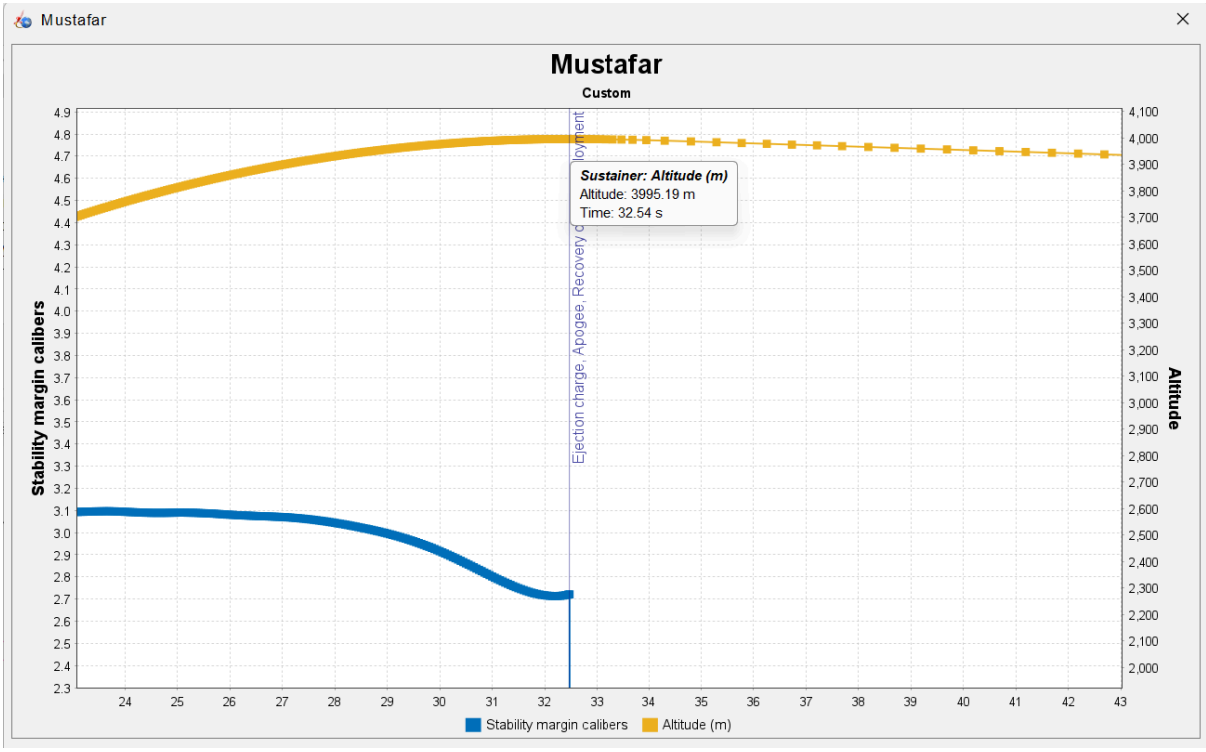

Mustafar
Mustafar
Custom
Motor burnout
Stage separation
Motor ignition
Motor burnout
Stability margin calibers
Altitude
Sustainer: Altitude (m)
Altitude: 364.56 m
Time: 2.94 s
Stability margin calibers
Altitude (m)

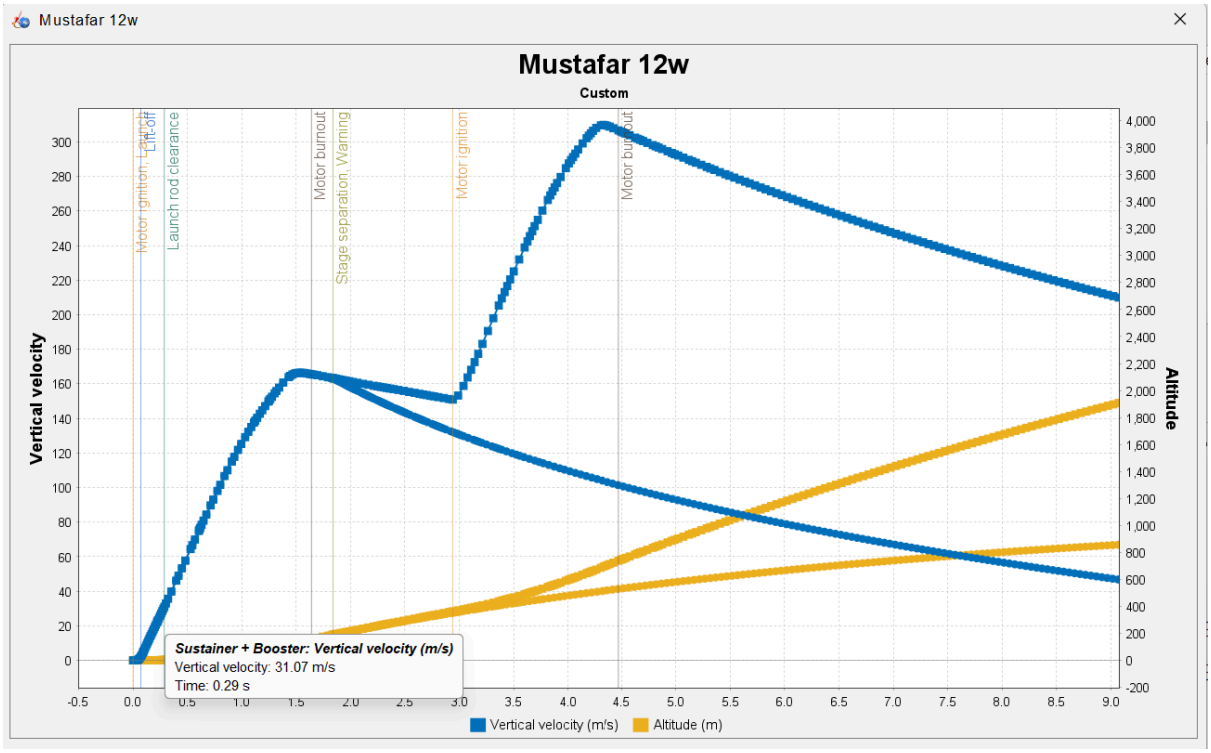

Mustafar
Mustafar
Custom
Motor ignition
Motor burnout
Stability margin calibers
Altitude
Sustainer: Stability margin calibers
Stability margin calibers: 3.1
Time: 4.49 s
Stability margin calibers
Altitude (m)

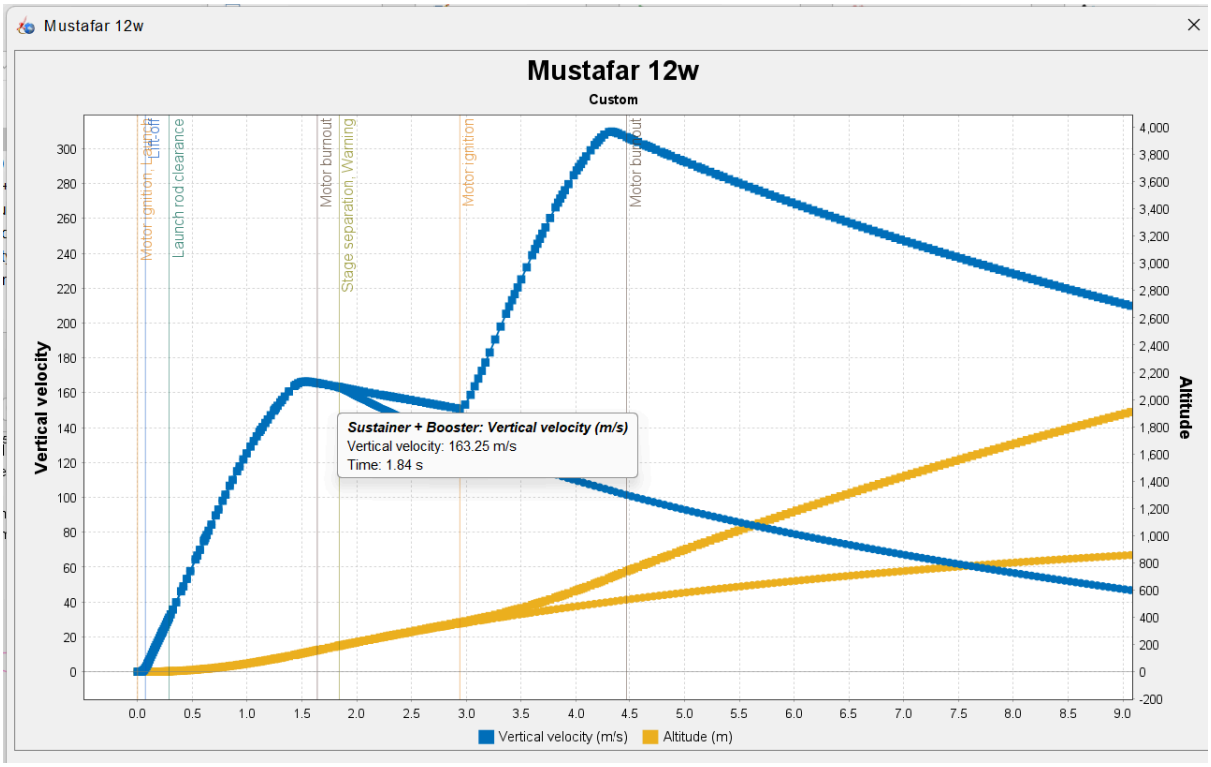

Mustafar
Mustafar
Custom
Motor burnout
Sustainer: Stability margin calibers
Stability margin calibers: 3.13
Time: 7 s
Stability margin calibers
Altitude
Stability margin calibers
Altitude (m)

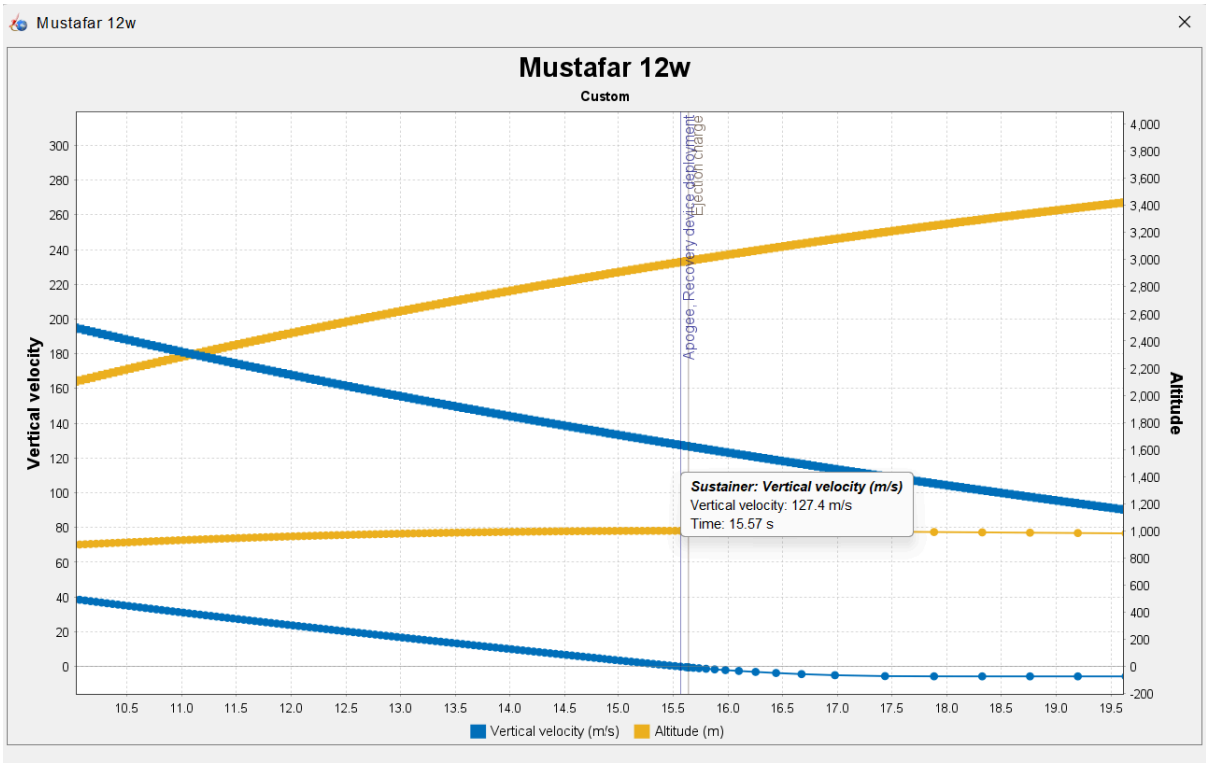

Mustafar
Mustafar
Custom
Motor ignition
Motor burnout
Sustainer: Altitude (m)
Altitude: 747.53 m
Time: 4.49 s
Stability margin calibers
Altitude
Stability margin calibers
Altitude (m)

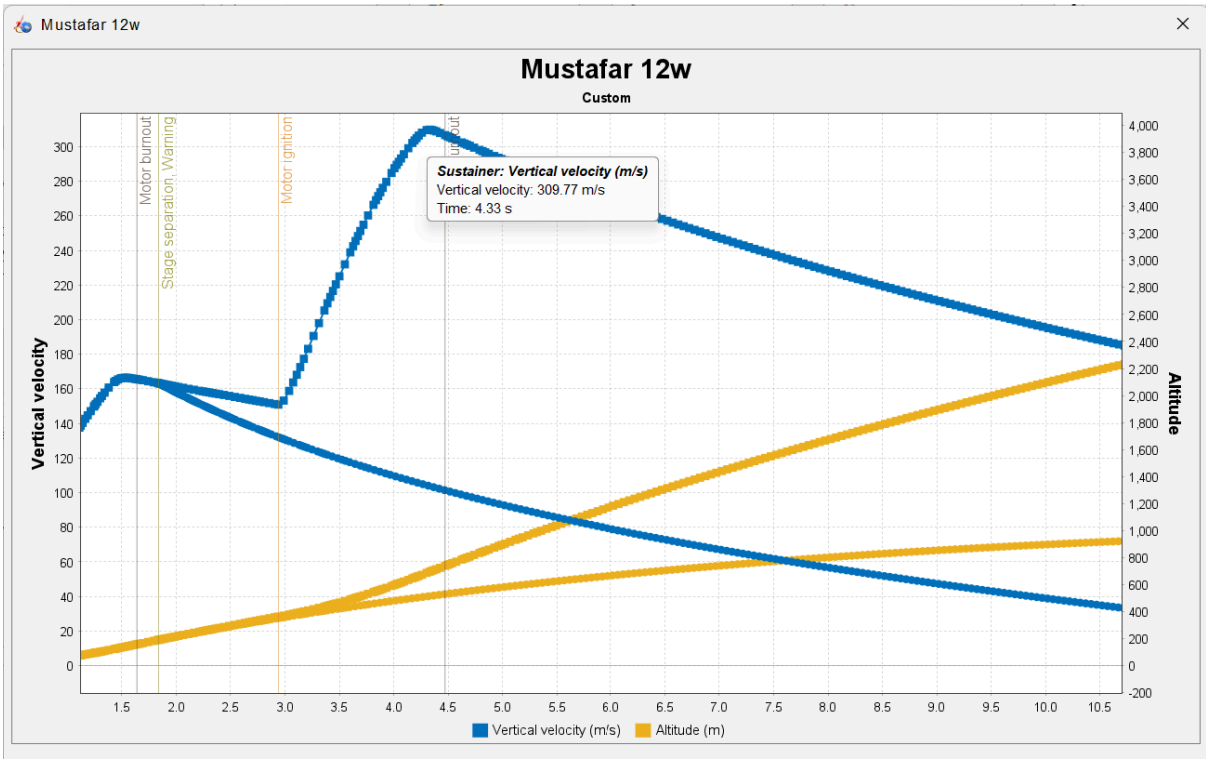

Mustafar
Mustafar
Custom
Stability margin calibers
Altitude
Ejection charge, Apogee, Recovery device deployment
Sustainer: Stability margin calibers
Stability margin calibers: 2.72
Time: 32.47 s
Stability margin calibers
Altitude (m)

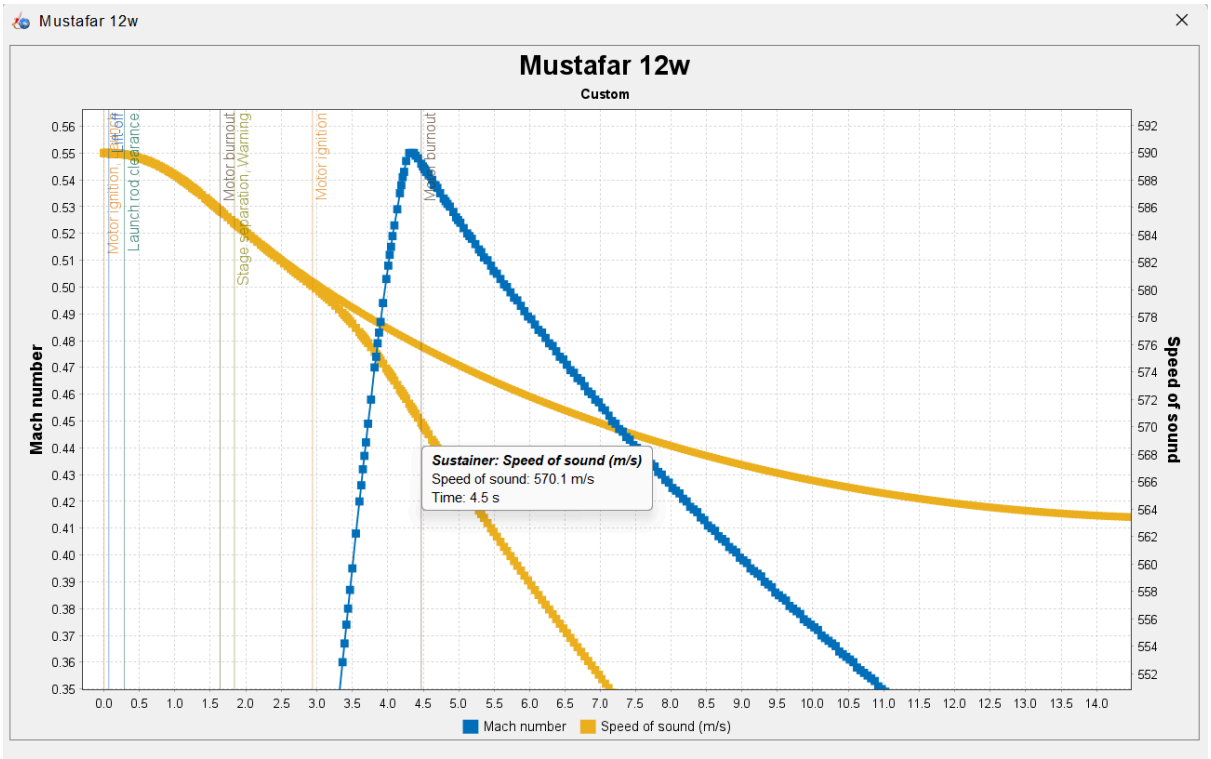

Mustafar
Mustafar
Custom
Stability margin calibers
Altitude
Sustainer: Altitude (m)
Altitude: 3995.19 m
Time: 32.54 s
Stability margin calibers
Altitude (m)

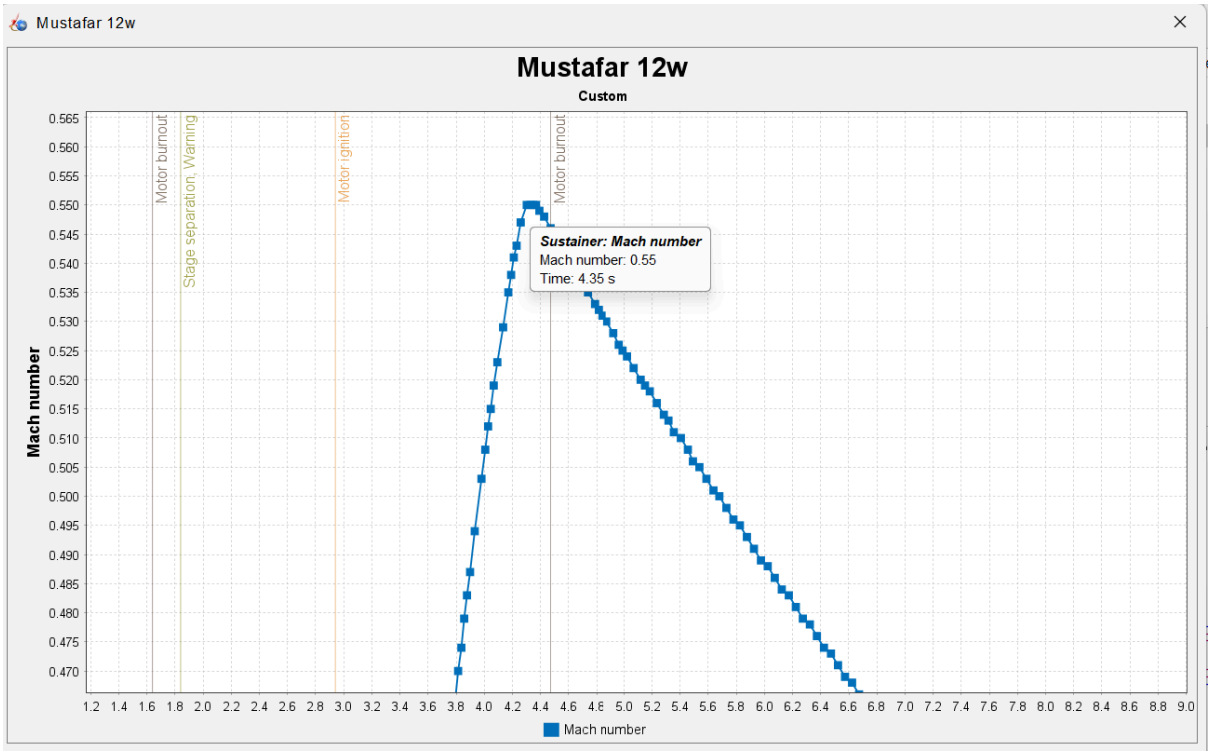

Mustafar 12w
Mustafar 12w
Custom
Motor ignition, Launch
Lift-off
Launch rod clearance
Motor burnout
Stage separation, Warning
Motor ignition
Motor burnout
Vertical velocity
Altitude
Sustainer + Booster: Vertical velocity (m/s)
Vertical velocity: 31.07 m/s
Time: 0.29 s
Vertical velocity (m/s)
Altitude (m)


Mustafar 12w
Mustafar 12w
Custom
Motor ignition, Launch
Lift-off
Launch rod clearance
Motor burnout
Stage separation, Warning
Motor ignition
Motor burnout
Vertical velocity
Altitude
Sustainer + Booster: Vertical velocity (m/s)
Vertical velocity: 163.25 m/s
Time: 1.84 s
Vertical velocity (m/s)
Altitude (m)


Mustafar 12w
Mustafar 12w
Custom
Apogee, Recovery device deployment
Ejection Charge
Sustainer: Vertical velocity (m/s)
Vertical velocity: 127.4 m/s
Time: 15.57 s
Vertical velocity
Altitude
Vertical velocity (m/s)
Altitude (m)


Mustafar 12w
Mustafar 12w
Custom
Motor burnout
Stage separation, Warning
Motor ignition
Sustainer: Vertical velocity (m/s)
Vertical velocity: 309.77 m/s
Time: 4.33 s
Vertical velocity
Altitude
Vertical velocity (m/s)
Altitude (m)


Mustafar 12w
Mustafar 12w
Custom
Motor ignition
Launch rod clearance
Motor burnout
Stage separation; Warning
Motor ignition
Motor burnout
Sustainer: Speed of sound (m/s)
Speed of sound: 570.1 m/s
Time: 4.5 s
Mach number
Speed of sound
Mach number
Speed of sound (m/s)


Mustafar 12w
Mustafar 12w
Custom
Motor burnout
Stage separation; Warning
Motor ignition
Motor burnout
Sustainer: Mach number
Mach number: 0.55
Time: 4.35 s
Mach number
Mach number